

FSD - Lecture 3 Notes

DOUBT LINK :

https://docs.google.com/forms/d/e/1FAIpQLScVNIFt_3for9LMQ59-VlicgS9qUbMo3f1wFkiBrkoayjPBfw/viewform?usp=publish-editor

1. The Real DOM (Document Object Model)

The Real DOM is the browser's internal map of your webpage.

Structure: It's a tree of objects representing every HTML tag.

The Problem: Direct manipulation is **expensive**. When you change one small thing (like a text color), the browser often has to recalculate the entire layout, styles, and "repaint" the screen.

Analogy: Imagine a massive notebook. To fix one typo on page 50, the browser has to flip through every page to make sure nothing else shifted.

2. The Virtual DOM (VDOM)

The Virtual DOM is a **lightweight, in-memory copy** of the Real DOM. It is a simple JavaScript object that looks like the real thing but has no power to change the screen directly.

How it Works (The 3-Step Process)

1. **Initial Snapshot:** When the app loads, React creates a Virtual DOM tree.
 2. **The Change:** When state changes (e.g., a counter increases), React creates a **new** Virtual DOM tree representing the update.
 3. **Diffing & Reconciliation:** React compares the new tree with the old one (this is called "Diffing"). It finds the *exact* differences and updates **only** those specific parts of the Real DOM.
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3. Comparison Table: Virtual DOM vs. Real DOM

Feature	Real DOM	Virtual DOM (VDOM)
Update Speed	Slow (Triggers full layout/repaint)	Fast (Only updates changed nodes)
Memory Usage	High (Tied to browser engine)	Efficient (Stored as a JS Object)
HTML Impact	Can produce "jank" if many items change	Minimal impact due to batching
Manipulation	Manual (using <code>getElementById</code>)	Automatic (managed by Framework)
Re-rendering ¹³	Re-renders the entire element & children ¹⁴	Re-renders only the "diffed" part ¹⁵

4. Why Use VDOM? (Key Benefits)

- **Batching:** Instead of updating the Real DOM 10 times for 10 changes, the VDOM collects them all and performs **one single update**.¹⁶
 - **Declarative Programming:** You tell the VDOM *what* you want the UI to look like (e.g., `count: 5`), and the VDOM handles the complex *how* of updating the browser.¹⁷
 - **Performance on Large Apps:** In simple websites, the Real DOM is fine.¹⁸ In massive apps (like Facebook or Gmail), the VDOM is essential to prevent the browser from freezing during constant updates.
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